

SVKMs NMIMS
MUKESH PATEL SCH.OF TECH.MGMT ENGG-NM-M

Programme: B. Tech (AI)/B. Tech (AI & ML)/B. Tech (COMP)/B. Tech (CS)/B. Tech (CSE - CS)/B. TECH (CSE - DS)/B. Tech (DS)/B. Tech (EXTC)/B. Tech Intg. (CE)/B. Tech Intg. (DS)/B. Tech Intg. (IT)/B. Tech Intg. (Mech)/B. Tech (MECH)/B. Tech (MXTC)

Year: III / V/Semester IX/Semester V(Exam Year: 2025-2026)

Subject: Elements of Project Management

Date: 11 Dec 2025

Time: 02:00 pm to 05:00 pm (03:00 Hrs.)

Max. Marks: 100

FINAL EXAMINATION (Acad. Year:2025-2026)

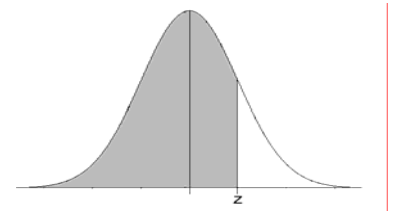
Instructions:

1. This question paper contains 4 pages.
2. Answer to each new question to be started on a fresh page.
3. Figure in right hand side indicates full marks
4. Question No. 1 is compulsory
5. Out of remaining questions, attempt any 4 questions
6. In all 5 questions to be attempted
7. All questions carry equal marks
8. Standard Normal Cumulative Probability Table attached
9. Assume suitable data if necessary

Q1		Answer briefly:	[20]
CO-1 ; SO-4 ; BL-3	a.	List the parameters based on which classification of projects can be done	[04]
CO-1 ; SO-5 ; BL-2	b.	Describe advantages of Matrix organization structure when one studies project management	[04]
CO-2 ; SO-4 ; BL-3	c.	Explain RACI-Matrix	[04]
CO-3 ; SO-4 ; BL-2	d.	Describe importance of cost estimation in project management	[04]
CO-3 ; SO-4 ; BL-3	e	Define ‘Project risk management’ and list three examples of risks in project management	[04]
Q2 CO-1; SO-4; BL-4	a	Explain significance of triple constraint along with quality as fourth constraint in project management	[10]
Q2 CO-1; SO-4; BL-4	b	Define ‘Project Manager’ also Analyze comprehensively the importance and significance of emotional intelligence of project manager on success and outcomes of project	[10]
Q3	a	“Gantt chart developed in 1918, still today it is important tool in project planning & scheduling” Explain in detail and also list its disadvantages	[10]

CO-2,4; SO-; BL-3																																			
Q3 CO-3; SO-4; BL-4	b	Explain Decision Tree Analysis as technique of quantitative risk analysis in project risk management	[10]																																
Q4 CO-2; SO-1; BL-5	a	The information about project is given in following table: <table border="1"> <thead> <tr> <th>Activity</th><th>To (weeks)</th><th>Tm (weeks)</th><th>Tp (weeks)</th></tr> </thead> <tbody> <tr> <td>1-2</td><td>1</td><td>1</td><td>7</td></tr> <tr> <td>1-3</td><td>1</td><td>4</td><td>7</td></tr> <tr> <td>1-4</td><td>2</td><td>2</td><td>8</td></tr> <tr> <td>2-5</td><td>1</td><td>1</td><td>1</td></tr> <tr> <td>3-5</td><td>2</td><td>5</td><td>14</td></tr> <tr> <td>4-6</td><td>2</td><td>5</td><td>8</td></tr> <tr> <td>5-6</td><td>3</td><td>6</td><td>15</td></tr> </tbody> </table> 1. Sketch network diagram 2. Evaluate critical path 3. Determine standard deviation and variance for each activity 4. Determine the probability that project will be completed no more than 4 weeks later than expected time 5. If the project due date is 19 weeks determine the probability of not meeting the due date 6. What should be the scheduled completion time for the probability of completion to be 90%?	Activity	To (weeks)	Tm (weeks)	Tp (weeks)	1-2	1	1	7	1-3	1	4	7	1-4	2	2	8	2-5	1	1	1	3-5	2	5	14	4-6	2	5	8	5-6	3	6	15	[12]
Activity	To (weeks)	Tm (weeks)	Tp (weeks)																																
1-2	1	1	7																																
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2-5	1	1	1																																
3-5	2	5	14																																
4-6	2	5	8																																
5-6	3	6	15																																
Q4 CO-3; SO-4; BL-3	b	Write in detail about ‘Identify Risks’ as one of seven processes in project risk management	[08]																																
Q5 CO-1; SO-4; BL-3	a	Sketch and explain System Development Cycle with perspectives of project management	[10]																																
Q5 CO-1; SO-4; BL-4	b	Explain detail procedure to construct WBS- Work Breakdown Structure (support your explanation with example)	[10]																																
Q6 CO-3; SO-4;	a	“Risk Register is important and significant document in project risk management” Explain	[08]																																

BL-4			
Q6 CO-3; SO-4; BL-2	b	Compare top-down cost estimation method with bottom-up cost estimation method	[04]
Q6 CO-3; SO-4; BL-3	c	Explain 'Project life cycle cost estimation'	[08]
Q7 CO-4; SO-4; BL-2	a	Discuss project termination and closeout responsibilities	[08]
Q7 CO-1; SO-5; BL-4	b	Compare Traditional(classical) organization structure with Projectized (pure product) organization structure	[04]
Q7 CO-3; SO-4; BL-3	c	Define 'Summary evaluation' and explain Project formative evaluation	[08]



Cumulative probabilities for POSITIVE z-values are shown in the following table:

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998